

Lesson 6-15: Simplify Algebraic Expressions

Grade 6 Mathematics · Standard 6.AT.7

Name: _____ Date: _____

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

Warm up with the lesson's core skills — one best answer each.

Question 1 SELECTED RESPONSE · 1 POINT

Simplify $7x + 3x$.

- ☐ A $10x$
- ☐ B $10x^2$
- ☐ C $21x$
- ☐ D $73x$

Question 2 SELECTED RESPONSE · 1 POINT

Simplify $9n + 4 + 2n$.

- ☐ A $11n + 4$
- ☐ B $11n^2 + 4$
- ☐ C $15n$
- ☐ D $92n + 4$

Question 3 SELECTED RESPONSE · 1 POINT

Which pair are like terms?

- ☐ A $6x$ and $2x$
- ☐ B $6x$ and $6y$
- ☐ C $6x$ and 6
- ☐ D $6x$ and $2x^2$

Question 4 SELECTED RESPONSE · 1 POINT

Simplify $5a + 3 + 2a$.

- ☐ A $7a + 3$
- ☐ B $7a^2 + 3$
- ☐ C $10a$
- ☐ D $52a + 3$

Question 5 SELECTED RESPONSE · 1 POINT

The board shows $3x + 5x$ on one channel. Simplify it.

- ☐ A $8x$
- ☐ B $8x^2$
- ☐ C $15x$
- ☐ D $35x$

Question 6 SELECTED RESPONSE · 1 POINT

Simplify the track: $4a + 2 + a$.

- ☐ A $5a + 2$
- ☐ B $5a^2 + 2$
- ☐ C $6a$
- ☐ D $7a$

Part B — Test-Format Questions

These match the state test's formats: two-part questions, select-all, and written responses.

Question 7**TWO-PART QUESTION****Part A**

Simplify the expression $8x + 3 + 2x - 1$.

Fill in the bubble next to the one best answer.

- ☐ (A) $6x + 2$
- ☐ (B) $10x + 2$
- ☐ (C) $10x + 4$
- ☐ (D) $12x$

Part B

Answer Part A before Part B — Part B asks about your reasoning.

Which reasoning shows why your answer to Part A is correct?

- ☐ (A) $8x$ and $2x$ share the same variable part, so add their coefficients to get $10x$. The constants 3 and -1 combine separately to 2 , giving $10x + 2$.
- ☐ (B) Combine every number in the expression into one coefficient: $8 + 3 + 2 - 1 = 12$, giving $12x$.
- ☐ (C) Subtract the coefficients of the variable terms: $8 - 2 = 6$, giving $6x + 2$.
- ☐ (D) Add the coefficients of the variable terms and then add both constants: $10x + 4$.

Question 8**SELECT ALL THAT APPLY**

Select ALL statements that are true about the expression $5y + 7 + 3y + y$.

Mark the box next to EVERY answer that is true. More than one is correct.

- ☐ (A) $5y$, $3y$, and y are like terms.
- ☐ (B) The coefficient of the term y is 1 .
- ☐ (C) The simplified expression is $9y + 7$.
- ☐ (D) The simplified expression is $16y$.
- ☐ (E) 7 is a constant and cannot be combined with the y terms.
- ☐ (F) The simplified expression is $8y + 7$.

Part C — Show Your Thinking

Answer in writing, the way the state test's constructed responses work.

Question 9

WRITTEN RESPONSE · 2 POINTS

Three students simplified $4x + 3 + 2x + 5 + x$ differently. Student A got $7x + 8$. Student B got $7x^2 + 8$. Student C got $6x + 9$. Who is correct? Explain the mistakes the other two students made.

Write your answer in complete sentences. Show or explain your mathematical thinking.

When you finish, check your reasoning with your teacher or family. Your teacher has the scoring guide for the written responses.